

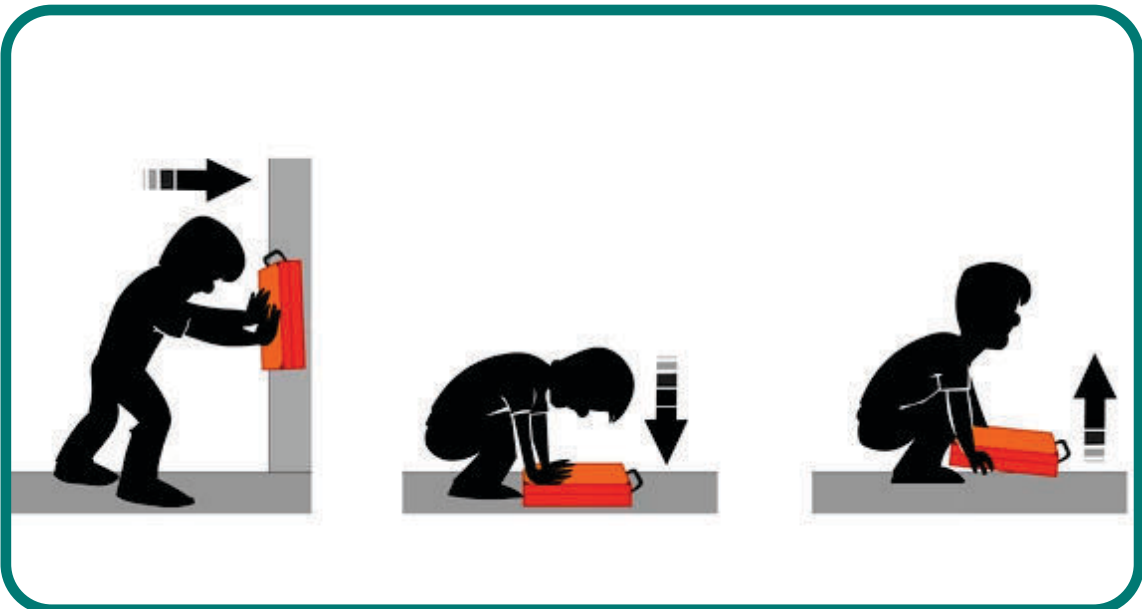
UNIT 8

FORCES AT WORK

Learning objectives:

1. Explain how simple machines make work easier.
2. Describe gravity's effect and compare weight on Earth and the Moon.
3. Understand magnetism, attraction, and repulsion.
4. Explain how sound is produced and varies in pitch/loudness.
5. Conduct simple experiments to observe forces in action.

THE BIG IDEA: WHAT CAN FORCES DO?



8.1 Tools and Machines

What makes work easier?

A **simple machine** makes moving an object easier. It makes the force of your push or pull stronger. A **lever** is a simple machine that lets you use less force to lift something. Some examples of levers are forks, scissors, and see-saws.



↑ You can use a metal stick as a lever to open a can of paint.

A **ramp** is a simple machine that helps you move things to a higher place. A wheel is another simple machine that helps you move things. Wheels let you roll objects instead of lifting them.



↑ Pushing the box up a ramp is easier than lifting it up.





↑ The wheels on the stroller make it easier for the girl to push it.

What is a tool?

A tool can be a simple machine, or it can be made up of many simple machines. A hammer is a tool that is one simple machine. An engine is a tool that is made up of many simple machines. All the parts work together to make a car move.



Look at these three tools.



↑ A spatula can pull or push food.



↑ A nutcracker creates a force strong enough to break hard shells.



↑ Turning the can opener's sharp wheel can cut a metal lid.

Quick Check

1. What does a hammer do?

2. What does a spatula do?



8.2 Gravity

What is gravity?

Gravity is a force that pulls things toward each other. The larger an object is, the stronger the force of gravity it has. You cannot see gravity, but you can feel it. Gravity keeps you on the ground.



↑ Gravity pulls the parachute toward Earth.

The Earth has a strong force of gravity because of its size. The gravity of Earth is stronger than the gravity of smaller things. That is why a ball in the air will fall back down to the Earth.



What is weight?

Gravity pulls things toward the center of the Earth with a certain amount of force. This amount of force is called **weight**. You can find out the weight of an object by putting it on a scale. A scale can measure weight in grams, kilograms, or even tons.



Planets and moons have more or less gravity than the Earth. Our Moon is much smaller than the Earth. That means it has less gravity than the Earth. Things weigh less on the Moon than on Earth. If you weighed 60 pounds on the Earth, you would weigh about 10 pounds on the Moon!



↑ An astronaut can jump high on the Moon because there is less gravity to pull him down.



8.3 MAGNETS PUSH, MAGNETS PULL

What does a magnet pull?

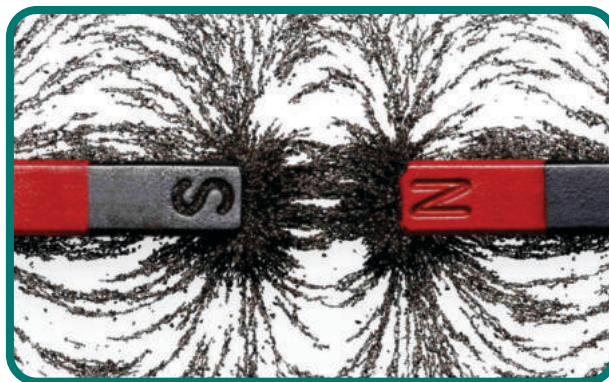
A magnet can **attract**, or pull, objects made of iron. A magnet will not attract a penny because it is not made of iron. Is a quarter made of iron? How could you find out? Magnets can move things without even touching them. They can pull through solids like paper, plastic, or glass. They can pull through liquids and gases, too.



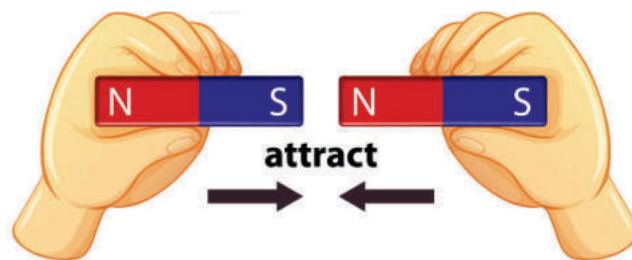
↑ Magnets can pull iron objects through liquids and solids.

What are poles?

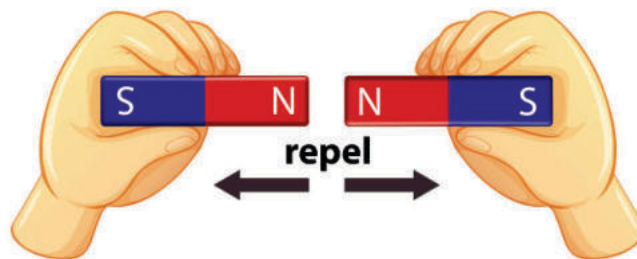
Magnets have two poles, a north pole and a south pole. The **poles** are where the pull of the magnet is the strongest.



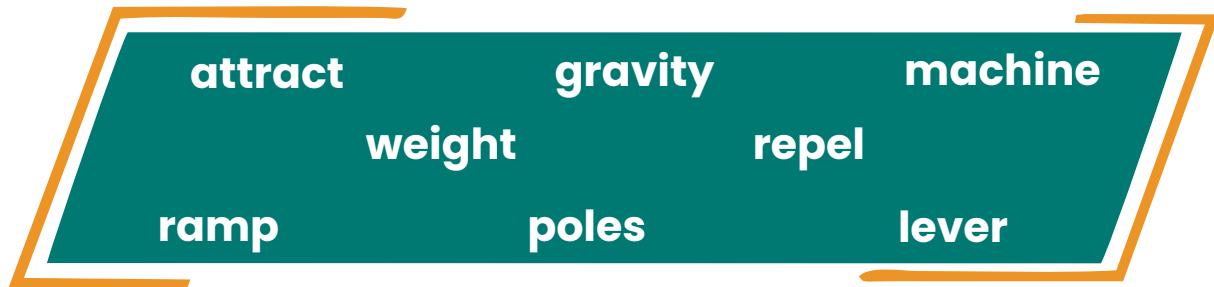
The poles of these magnets **attract** each other because they are opposites. A north pole and a south pole pull toward each other.



The poles of these magnets **repel**, or push away from each other. They repel each other because they are the same.



Write the number of the correct definition next to each word.



1. pull objects made of iron.
2. the amount of force an object has.
3. It lets you use less force to lift something.
4. A simple _____ makes the force of your push or pull stronger.
5. the two parts of a magnet where its pull is the strongest.
6. a force that pulls things toward each other.
7. when two poles of magnets push away from each other.
8. a machine that helps you move things to a higher place.



Test Your Understanding

Q1. How does a simple machine make work easier?

Ans: _____

Q2. What is gravity and how does it affect objects?

Ans: _____

Q3. How do magnets attract or repel objects?

Ans: _____

Q4. Why do objects weigh less on the Moon than on the Earth?

Ans: _____

Q5. What happens when the poles of two magnets are the same?

Ans: _____



Fill in the blanks.

1. A _____ is a tool that makes work easier by increasing force or changing direction.
2. Gravity pulls objects toward the _____ of the Earth.
3. A magnet will only attract objects made of _____.
4. The Moon has _____ gravity than the Earth.
5. A ramp is a type of _____ machine that helps move objects to a higher place.



Tick (✓) for True and cross (x) for False statements.

1. A lever is a type of simple machine that reduces () effort.
2. The force of gravity is the same on the Earth () and the Moon.
3. Magnets can attract all types of metals. ()
4. A hammer is not a simple machine. ()
5. Weight can be measured in metres. ()



Multiple Choice Question.

1. Which of the following is an example of a simple machine?

- a) Spoon
- b) Ramp
- c) Cup
- d) Chair

2. What will happen if you drop a ball on Earth?

- a) It will float
- b) It will remain in place
- c) It will fall to the ground
- d) It will move sideways

3. Why does a magnet attract an iron nail but not a wooden stick?

- a) Iron is shiny
- b) Magnets only attract iron-based objects
- c) Wood is heavier than iron
- d) Magnets attract all objects

4. What happens when you bring the north poles of two magnets together?

- a) They attract
- b) They repel
- c) They combine
- d) They do nothing





5. Which force keeps us from floating into space?

- a) Friction
- b) Gravity
- c) Magnetism
- d) Wind





Activity (Hands-On Experiment)



ACTIVITY

Magnet Power!

.Objective: Discover how magnets attract and repel.

.Materials: Two bar magnets, small iron nails, a plastic spoon, a piece of aluminum foil.

.Procedure:

1. Place the two magnets close to each other with opposite poles facing. Observe what happens.
2. Flip one magnet so the same poles are facing. Observe again.
3. Try picking up different materials (nails, spoon, foil) with the magnet.



Observation Questions:

- 1.What happens when opposite poles are facing?
- 2.What happens when similar poles face each other?
- 3.Which objects did the magnet attract? Why?

[illegible]

GLOSSARY

- **Environment:** all the living and non-living things in a place.
- **Biome:** a place with certain kinds of living and non-living things.
- **Adaptation:** a body part or way of acting that helps a living thing survive.
- **Desert:** a place with hot, dry weather.
- **Migrate:** to move from one place to another.
- **Grassland:** a place with many grasses.
- **Forest:** a place with many trees.
- **Mimicry:** when one thing pretends to be another thing.
- **Arctic tundra:** a cold place above the Arctic Circle.
- **pollution:** harmful things getting into the air, water, or land
- **reduce:** to use less of something.
- **reuse:** to use something again.
- **recycle:** to turn an old thing into something new.
- **habitat:** the place where a living thing makes its home.
- **extinct:** all dead, no more of that living thing alive on Earth.
- **fossil:** the hardened remains of a living thing.
- **geologist:** a scientist who studies rocks and puts them into groups.
- **minerals:** what rocks are made of.



- **property:** tells you something about an object.
For example, its colour.
- **lustre:** how a mineral looks when light shines on it
- **hardness:** how tough a mineral is.
- **weathering:** the way water and wind change rocks.
- **soil:** made up of tiny rocks and bits of plants and animals.
- **simple machine** – a tool that makes the force of your push or pull stronger.
- **lever** – a simple machine that lets you use less force to lift something.
- **ramp** – a simple machine that helps you move things to a higher place.
- **gravity** – a force that pulls things toward each other.
- **weight** – the amount of force an object has.
- **attract** – pull objects.
- **poles** – the two parts of a magnet where its pull is the strongest.
- **repel** – when the poles of two magnets push away from each other.



Bravo! You've explored the wonders of science! From life cycles to the solar system, rocks, soil, machines, and more—you are now a young scientist! Keep exploring and asking questions, because science is everywhere!

